

Logistic Regression Assignment

<https://github.com/Rizwanali324/Python/tree/main/dataset/classification>

Student Mental Health & Academic Performance Dataset

Section 1: Data Understanding

1. How many rows and columns are present in the dataset? Based on the sample provided, what is the total number of unique students?
 2. Which columns have missing values? Examine the provided sample are there any visible missing entries? How would you systematically check for missing values across the entire dataset?
 3. Identify all categorical, numerical, and ordinal columns in the dataset. Justify your classification for each.
 4. Calculate the mean, median, and standard deviation for **cgpa**, **daily_study_hours**, and **screen_time_hours**.
 5. Which gender has the highest average **anxiety_score**? Provide the exact value.
 6. List all unique **course** values and calculate the proportion of students in each course.
 7. What is the distribution of **year of study**? How many students are in each academic year?
 8. Compare the average **cgpa** between students with **sleep_quality** = "Good" vs "Poor".
 9. What is the correlation between **daily_study_hours** and **cgpa** in the sample?
 10. Which **stress_level** category appears most frequently, and what is its percentage?
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Section 2: Data Cleaning & Preprocessing (Complex)

Missing Value Handling

- a. How many rows contain at least one NaN value?
- b. If missing values exist, what strategies would you use for numerical columns vs categorical columns? Provide at least two different imputation methods for each type with justification.
- c. Are there any rows with all NaN values? How would you handle them?

Duplicate Detection

- a. How many duplicate rows exist? What columns would you consider to define a duplicate record?
- b. If duplicates are found, what effect might keeping them have on logistic regression model performance?

Outlier Detection & Treatment

- a. Use the IQR method to detect outliers. How many outliers exist in each?
- b. Apply two different outlier treatment methods (iqr and transformation) and explain which you would choose for logistic regression and why.

Encoding Categorical Variables

- a. Identify all categorical columns and explain why they need encoding.
- b. Apply both One-Hot Encoding and Label Encoding to **appropriate columns**. Compare the results.
- c. For **sleep_quality (Average, Good, Poor)** and **stress_level (Low, Medium, High)**, why is ordinal encoding more appropriate than one-hot encoding? Implement it.

Feature Scaling

- a. Explain why standardization (Z-score) is recommended for logistic regression.
- b. Apply StandardScaler to all numerical features. Show the mean and standard deviation of scaled features to verify.

Data Type Validation

After preprocessing, verify that all columns have appropriate data types for modeling. Convert any incorrectly typed columns.

Section 3: Exploratory Data Analysis (EDA)

1. Plot the distribution of **cgpa**. Is it normally distributed? Comment on skewness and kurtosis.
2. Plot a boxplot of **cgpa** by **stress_level**. What insights can you derive about the relationship between stress and academic performance?
3. Create a bar plot showing average **anxiety_score** by **gender**. Which gender shows higher anxiety, and does this align with common expectations?
4. Plot a scatter plot with **daily_study_hours** on x-axis and **cgpa** on y-axis, colored by **sleep_quality**. What patterns emerge?

5. Create a grouped bar chart showing the count of students in each **burnout_level** across different course types. Which course has the highest burnout proportion?
 6. Plot the distribution of **screen_time_hours**. Identify any extreme values and relate them to **depression_score**.
 7. Create a violin plot of **cgpa** by **internet_quality**. What does this suggest about internet quality's impact on academics?
 8. Plot a heatmap of the correlation matrix for all numerical variables. Which features are most correlated with **cgpa**? With **anxiety_score**?
 9. Create a pairplot for **daily_study_hours**, **screen_time_hours**, **cgpa**, and **anxiety_score**. Which relationships appear non-linear?
 10. Boxplot of **academic_pressure_score** by **burnout_level**. Is there a statistically significant difference visible?
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Section 4: Model Building — Binary Classification

Target Variable Creation

Create a binary target variable **high_burnout** where:

- 1 if **burnout_level** = "High"
- 0 if **burnout_level** = "Low" or "Medium"

Explain why you chose this binarization.

Feature Selection

- a. Which columns will you use as features (X) for prediction? Justify your selection.
- b. Which column is the target (y)?
- c. Do you need to drop any columns to prevent data leakage? Identify them.

Train-Test Split

- a. Split the data into training and testing sets with 80–20 ratio and **random_state = 42**. Why is **random_state** important?
- b. What is the purpose of splitting the dataset before training?

Model Initialization & Training

- a. Initialize a Logistic Regression model with **max_iter = 1000**. Why might you need to increase **max_iter**?
- b. Fit the model on the training data. What is the training accuracy score?

Section 5: Model Evaluation

Predictions & Metrics

- a. Predict on the test set and calculate accuracy, precision, recall, and F1-score.
- b. Generate and interpret the confusion matrix.

ROC-AUC Analysis

- a. Plot the ROC curve and calculate the AUC score. What does this tell you about model discriminative ability?
- b. If AUC is below 0.7, what might be the reasons?

Classification Report Analysis

Provide and interpret the classification report. Which class (**high_burnout = 0 or 1**) is the model better at predicting? Why might this be happening?

Section 6: Model Interpretation & Critical Thinking

Coefficient Analysis

- a. Extract and interpret the coefficients of the logistic regression model. Which features have the strongest positive and negative influence on predicting high burnout?
- b. How would you explain the impact of **academic_pressure_score** on burnout odds to a non-technical stakeholder?

Class Imbalance Handling

- a. Check the class distribution of **high_burnout**. Is there imbalance?
- b. If imbalanced, apply SMOTE oversampling and retrain the model. Compare the new performance metrics with the original model.

Preprocessing Impact Assessment

- a. Train the model again without feature scaling. Compare coefficients and performance. Why does scaling matter for logistic regression?
- b. Train the model without outlier treatment. How does performance change?

Model Improvement Strategies

- a. If you wanted to improve model accuracy further, what additional steps could you take? Propose at least 3 specific strategies with justification.
- b. Try L1 (Lasso) regularization and compare feature selection with your original model. Which features become zero coefficients?

Cross-Validation

Perform 5-fold cross-validation on the logistic regression model. Report mean accuracy and standard deviation. What does the standard deviation indicate about model stability?

Alternative Algorithm Comparison

Train a Random Forest Classifier on the same data. Compare its performance (accuracy, precision, recall) with logistic regression. Which performs better and why might that be the case?

Submission Requirements

- Submit a **Jupyter Notebook** with all code, outputs, and markdown explanations.
 - Each section should have **clear headings**.
 - Include comments explaining **critical decisions**.
 - Provide a **summary paragraph** at the end discussing key findings and model limitations.
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